TEXAS STATE UNIVERSITY

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University Nanosatellite Program MC
Final Event Presentation

ADAPT

[Adaptive Drag-Aware Pointing Testbed]

Jasmine Rawles Saidi Adams Ariana Rider

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Mission Overview

Mission Narrative

Satellites operating in Very Low Earth Orbit (VLEO) experience increased atmospheric drag that is further amplified during space weather events, accelerating orbital decay and degrading attitude pointing performance. Space weather forecasting can predict such events minutes to days in advance, limiting the time available for mission operators to plan and uplink spacecraft operational updates before storm conditions develop. Additionally, classical attitude controllers used are tuned on the ground and cannot adapt autonomously to the rapidly changing disturbances. ADAPT seeks to test the performance of a dynamic ML-based controller in comparison to a classical static controller in this environment.

Military and NASA Relevance

Autonomous Spacecraft Guidance, Navigation, and Control (GNC)

- Demonstrates and tests ML-enabled attitude control, highlighting areas of improvement and increasing spacecraft autonomy
- Assesses GNC performance in highly dynamic environments by adapting to changing atmospheric disturbances

Resource-Efficient and Resilient Spacecraft Operations

- Generates quantitative performance metrics for evaluating autonomous control system efficiency
- Identifies tradeoffs between pointing performance and resource utilization to inform future spacecraft control systems

VLEO Characterization and Awareness

- Characterizes VLEO disturbances by measuring changes in spacecraft pointing performance as environmental conditions vary throughout the mission

Mission Statement and Objectives

	REF	DESCRIPTION	PARENT REF
Mission Statement	MS	ADAPT will compare CubeSat pointing performance metrics between an ML-based attitude controller and a classical baseline controller in the VLEO environment.	
Mission Objectives	MO-1	Demonstrate pointing control of satellite using an ML-based attitude controller.	MS
	MO-2	Demonstrate the ability to update the ML-based attitude controller on-board.	MS
	MO-3	Compare the three-axis pointing RMS error between an ML-based controller and the classical controller.	MS
	MO-4	Compare actuator effort between an ML-based attitude controller and a classical controller.	MS

Success Criteria Metrics

- Pointing Performance Index
  - Accounts for three-axis RMS error, attitude jitter, and pointing deviation to compare error of the ML-based controller and the classical controller

$$PPI_W = \frac{E_{W,ML}}{E_{W,PID}}, \qquad E = (w_1 \cdot \theta_{RMS}) + (w_2 \cdot \omega_{RMS}) + (w_3 \cdot \theta_{max})$$

Actuator Performance Index

- Accounts for electrical power draw, torque RMS, momentum accumulation rate, and desaturation frequency to compare the ML-based controller and the classical controller

$$API_W = \frac{J_{W,ML}}{J_{W,PID}}, \qquad J = (w_1 \cdot \bar{P}) + (w_2 \cdot \bar{\tau}_{RMS}) + (w_3 \cdot \dot{H}_{accum}) + (w_4 \cdot N_{desat})$$

Success Criteria Metrics

	ORBITAL REGIME	DESCRIPTION	KP (AP) INDEX
Space Weather	QUIET	<table><tr><th></th><th>ORBITAL REGIME</th><th>DESCRIPTION</th><th>KP (AP) INDEX</th></tr><tr><td rowspan="2">Space Weather</td><td>QUIET</td><td>F10.7 solar flux of 65 to 110 SFU. Stable baseline atmospheric density of roughly [$1.5 \times 10^{-13}\ kg/m^3$, $1.0 > m^3$] [TBR].</td><td>0-4 (0-39)</td></tr><tr><td>STORM</td><td>Storm Mode prioritized. F10.7 solar flux of 150 to 250+ SFU. Features a 4-8x expansion in localized atmospheric density.</td><td>5-9 (56-400)</td></tr><tr><td rowspan="2">Illumination</td><td>SUNLIGHT</td><td>CubeSat in sight of the Sun.</td><td>-</td></tr><tr><td>ECLIPSE</td><td>CubeSat is not in sight of the Sun.</td><td>-</td></tr></table>	0-4 (0-39)
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Illumination	SUNLIGHT	CubeSat in sight of the Sun.	-
	ECLIPSE	CubeSat is not in sight of the Sun.	-

Success Criteria

	REF	DESCRIPTION	PARENT REF
Minimum Success Criteria	MSC-1	Quantify the Pointing Performance Index (PPI), for at least 2 [TBR] orbital regime within a 50 [TBR] km altitude band.	MO-1, MO-3
	MSC-2	Quantify the Actuator Performance Index (API), for at least 2 [TBR] orbital regime within a 50 [TBR] km altitude band.	MO-1, MO-4
Full Success Criteria	FSC-1	ML-based controller demonstrates reduction in pointing error, quantified as a [TBD] % decrease in the internal loss function relative to its initial on-orbit performance.	MO-2
	FSC-2	Quantify the Pointing Performance Index (PPI), for at least 3 [TBR] orbital regimes.	MO-3
	FSC-3	Quantify the Actuator Performance Index (API), for at least 3 [TBR] orbital regimes.	MO-4
Extended Success Criteria	ESC-1	Quantify Pointing Performance Index (PPI) to be PPI < (1 - ME) (i.e., pointing RMS error less than or equal to the classical PID controller) in at least 1 [TBR] orbital regime.	MO-3
	ESC-2	Quantify Actuator Performance Index (API) to be API ≤ (1 - ME) (i.e., pointing RMS error less than or equal to the classical PID controller) in at least 1 [TBR] orbital regime.	MO-4

Mission Constraints/Key Drivers/Requirements

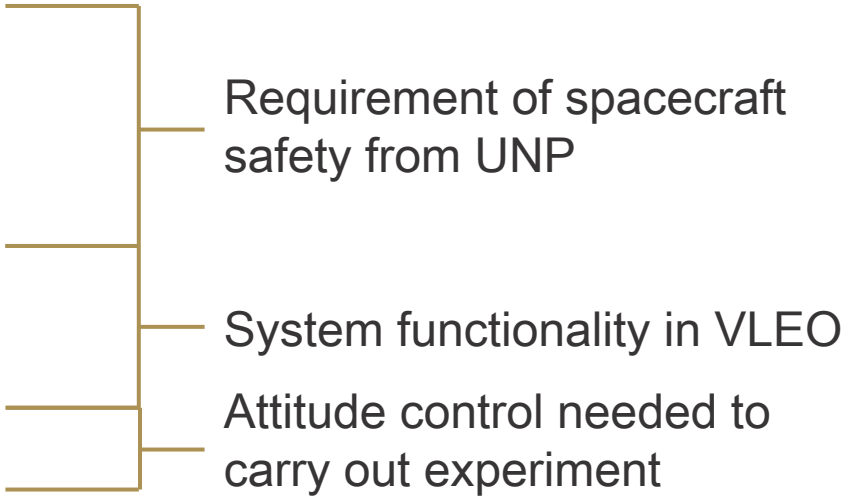
Constraints

- Must be a CubeSat
- Must communicate with TXST ground station

Drivers

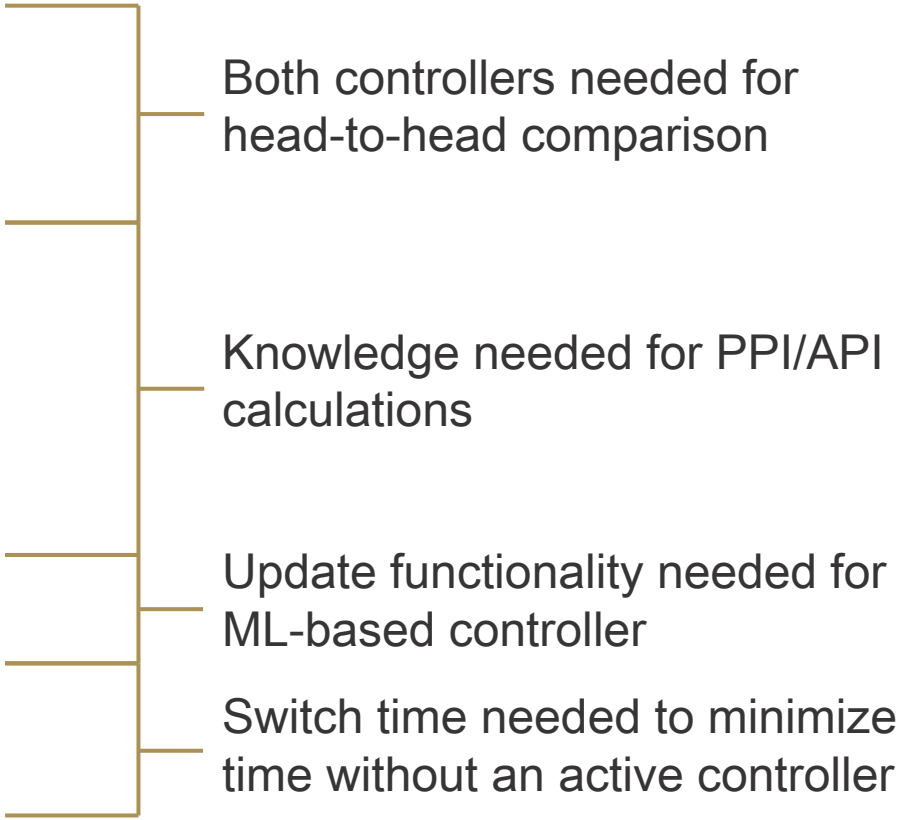
- Low cost
- Use of ML-based control

Number	Requirement
SYS-1	Spacecraft shall be designed to withstand the launch and on-orbit environments of the launch vehicle without failure that results in damage of launch vehicle and its contents or failure that causes injury to the ground handling crew.
SYS-2	The ADAPT system shall operate within a VLEO altitude range below 450km [TBR] for the mission duration of 360 days [TBR].
SYS-3	The ADAPT system shall have three-axis attitude control.



Mission Constraints/Key Drivers/Requirements

Number	Requirement
SYS-4	The ADAPT system shall execute a ML-based attitude control algorithm onboard.
SYS-5	The ADAPT system shall execute a classical PID attitude controller onboard.
SYS-6	The ADAPT system shall maintain attitude knowledge accuracy of ≤ [TBD] degree RMS (3-sigma) per axis.
SYS-7	The ADAPT system shall maintain positional knowledge within [TBD] meters (3-sigma).
SYS-8	The ADAPT system shall collect actuator usage data at a sampling rate of [TBD] Hz.
SYS-9	The ADAPT system shall be capable of updating on-board ML model parameters during mission.
SYS-10	The ADAPT system's payload and ADCS subsystems shall transition between operational modes within [TBD] ms of each other.



Experiment Plan

Payload Design

Together:

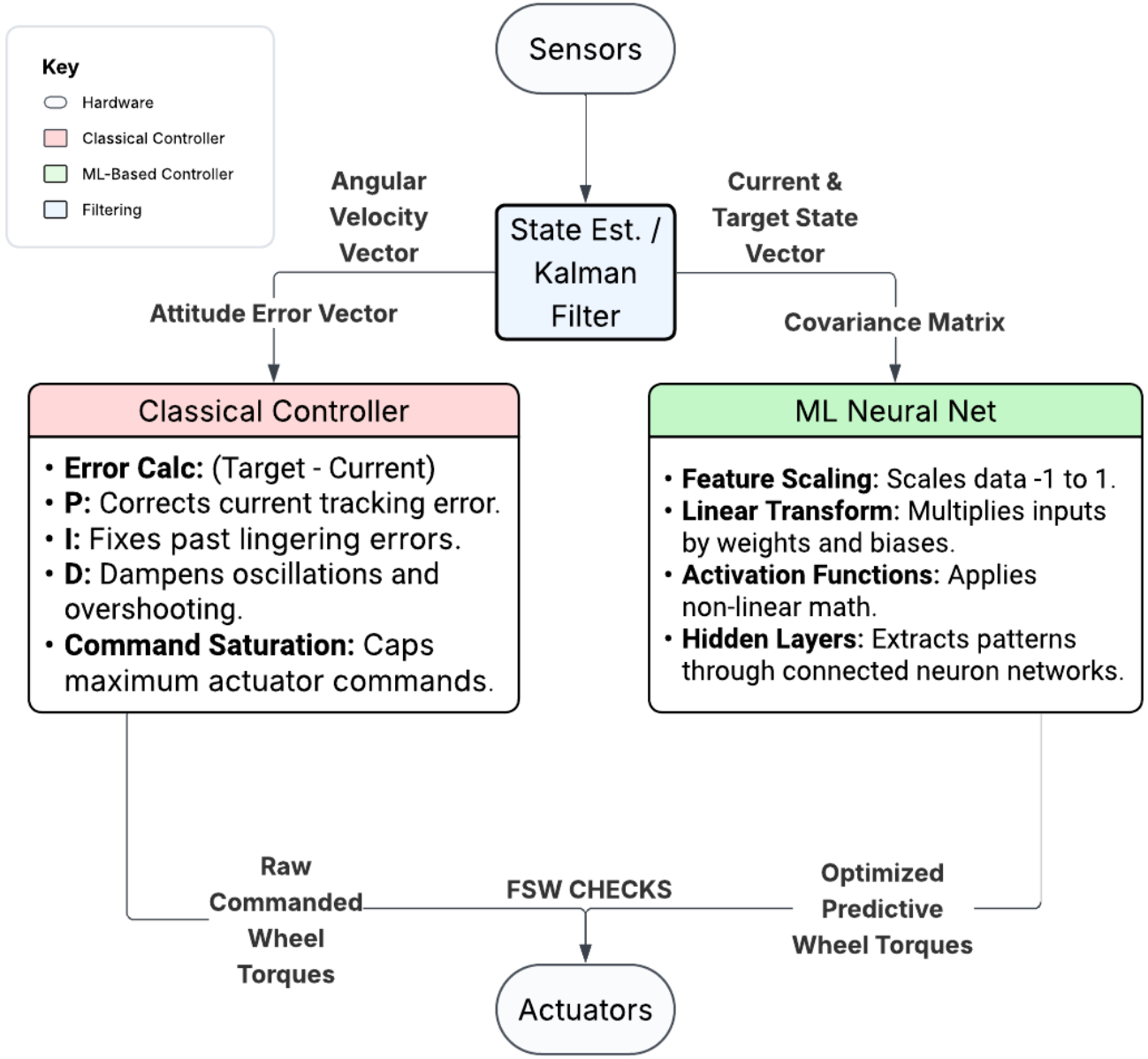
- Same sensor data & Actuator suite
- Different inputs from the KF pipeline
- Exact Classical Controller TBD (contingent on manufacturer and AoA)

COTS controller:

- On flight computer
- Reactive, linear (usually)

ML controller:

- On its own processor
  - NovoSpace GPU002SF card
- Predictive, non-linear; Adaptive
- Takes actuator usage into consideration



Orbit Simulation

Criteria	Coupled Orbit/Attitude Dynamics	VLEO Atmosphere & Drag Modeling	Component-Level Integration	Estimation (UKF) Out of the box	RL Pipeline Suitability & Speed	Open Source?/ Cost
GMAT	Low (Attitude is decoupled)	Good (MSISE-90, JR)	Low (Trajectory/ hardware decoupled)	Yes (Filter capabilities exist, but only EKF native)	Poor (Not built for parallelize step-loops, does have GMAT Python API)	Open Source Free
Basilisk	Excellent (Built for fully-coupled ADCS/GNC)	Great (Faceted/CAD drag, MSISE, Custom)	Excellent (Modular FSW, reaction wheels, sensors, power)	Yes (Native UKF modules for attitude/orbit)	Excellent (C++ core, Python wrapper, native bsk_rl library)	Open Source Free
STK	Moderate (Requires Solis/ODTK)	Excellent (JB2008, MSIS, Custom)	Moderate (Heavy UI/API wrapper)	Yes (via ODTK, expensive)	Poor (slow API, closed-source licensing)	Proprietary, Expensive
Orekit	Low (Primarily orbit-focused)	Excellent (JB2008, DTM, MSIS, DSST)	Low (Requires heavy custom Java coding)	Yes (EKF, ESKF, UKF, USKF)	Poor (Java-Python overhead, slow step speeds	Open Source Free

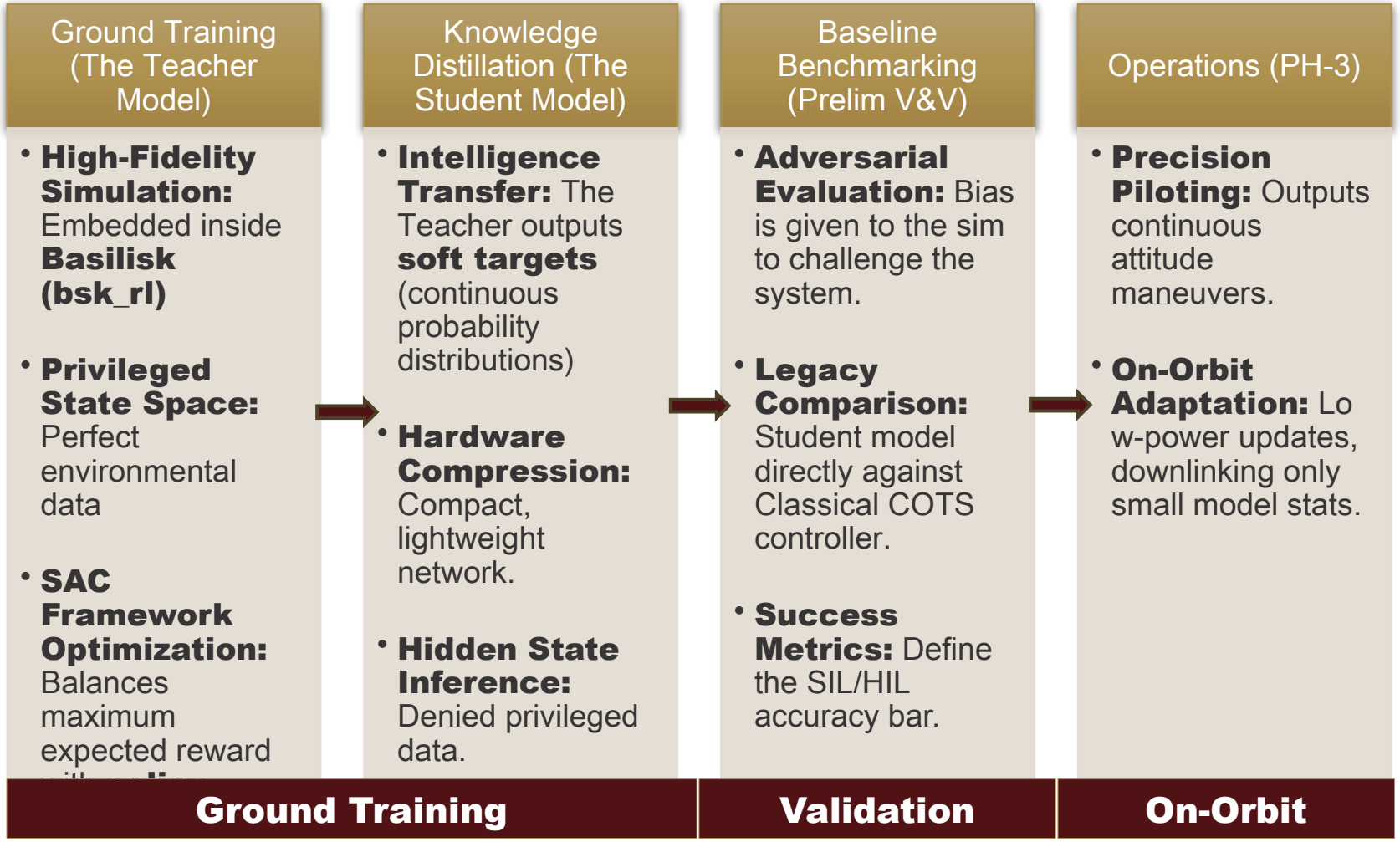
Why Basilisk?

- **High-Fidelity VLEO Physics:** Custom atmospheric models.
- **Built-In Machine Learning:** bsk_rl library.
- **Modular Architecture:** Mirrors FSW for HIL testing.
- **100% Code Reusability:** The exact same FSW algorithms written and tested in early simulations are compiled and run directly on in-flight hardware.

Mission Usage:

- **Pre-Launch Ground Training**
- **Benchmarking**
- **Trajectory estimation** (with integrated atmospheric models and geomagnetic/solar storm data)
    - Data for mission life, phases, and structural/component lifetimes
- **HIL Validation**

In-House Controller Pipeline



Concept:

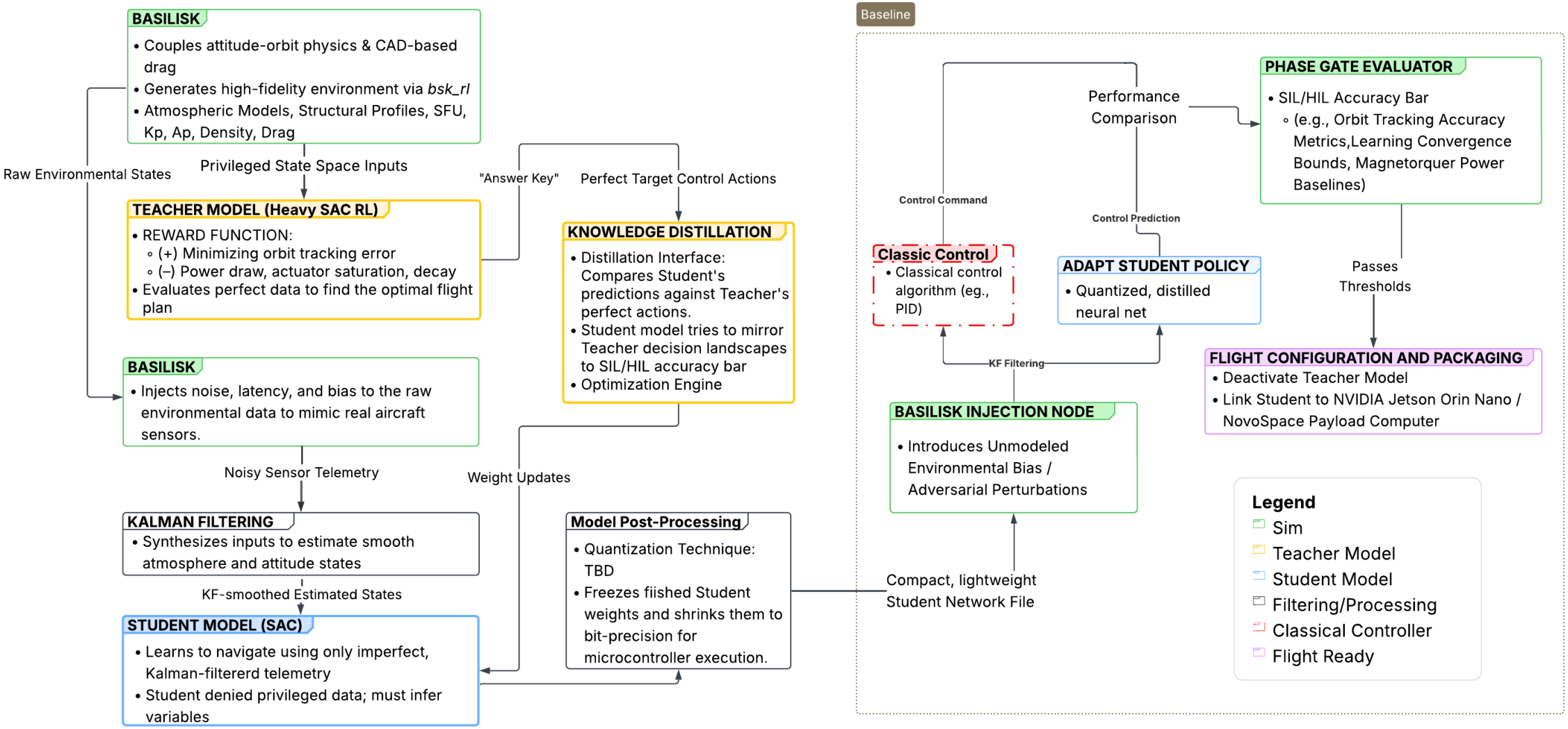
A Kalman/UKF estimator for state, density, and drag coefficient, feeding a learned control policy (compact with distilled pipeline built from teacher-student method). Combines classical estimation with learned nonlinear control. Control policy can further learn changing parameters on orbit.

On-Board Data Generation

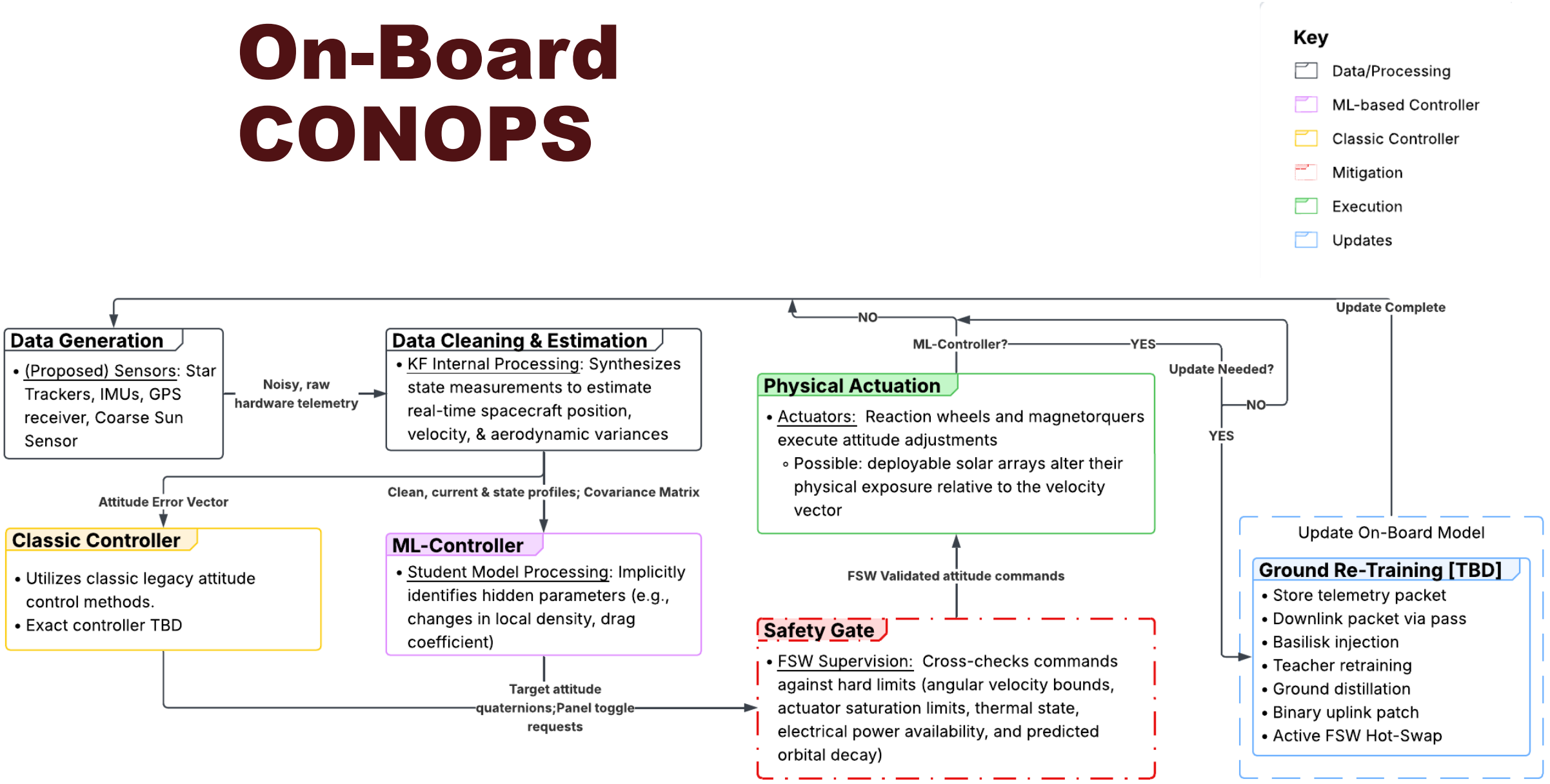
PPI	API
Required Hardware: - **Sensors**: Star Trackers (2+), Fine Sun Sensors (2-6), Inertial Measurement Units (IMUs), Microcontroller (2).	**Required Hardware:** - **COTS ADCS:** Reaction Wheels (3+), Magnetorquers (3+), Magnetometer (1), Motor, Hall Effect Sensors
Calculations: - Estimated Attitude = Current Orientation Quaternion ○ Source: Star Tracker - Target Attitude = Commanded Target Quaternion ○ Source: Microcontroller - Angular Velocity Error = Residual Body Rate ○ Source: IMUs - Sensor Health = Binary Status Flags ○ Source: Individual Sensor Health Checks	**Calculations:** - Electrical Power Draw = The Actuator Rail Current and Voltage ○ Source: Internal motherboard telemetry sensors from COTS ADCS system - RMS Torque = Commanded torque ○ Source: Microcontroller - Momentum Accumulation Rate = The wheel angular velocity or RPM. ○ Source: Motor's Hall Effect Sensors or Optical Encoders. - Desaturation Frequency = RW desaturation (dumping momentum) ○ Source: Reaction Wheels (RW)

CONOPS

Ground CONOPS



On-Board CONOPS



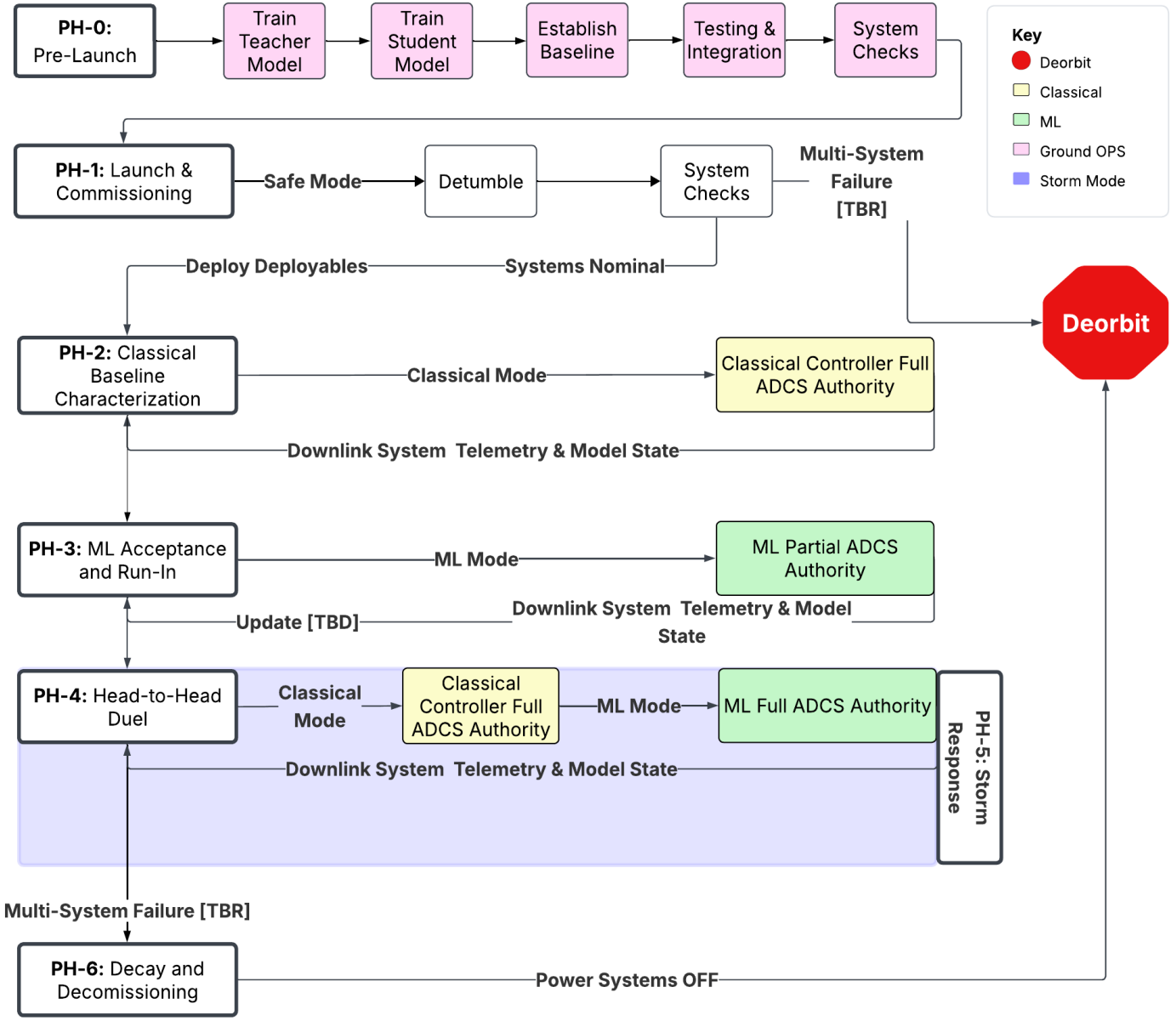
Mission Life CONOPS

Based on injection at 450km:

- PH-0 duration: TBD
- PH-1 duration: ~1-2 weeks
- PH-2 & PH-3 duration (combined): ~50km
  - (full drag ~75-95 days, least drag ~215-240 days)
- PH-4 duration: ~ 100km
  - (full drag ~35-50 days, least drag ~ 80-90 days)
- PH-5 duration: None Required
  - (weather based, contingent on Storm Regime metrics)

Total In-flight Expectation: ~132 – 388 days

Total Mission Expectation: TBD + (~132 – 388 days)



Orbit CONOPS





1. Command slew to 1st target to begin pointing

2. ML or Classical Mode

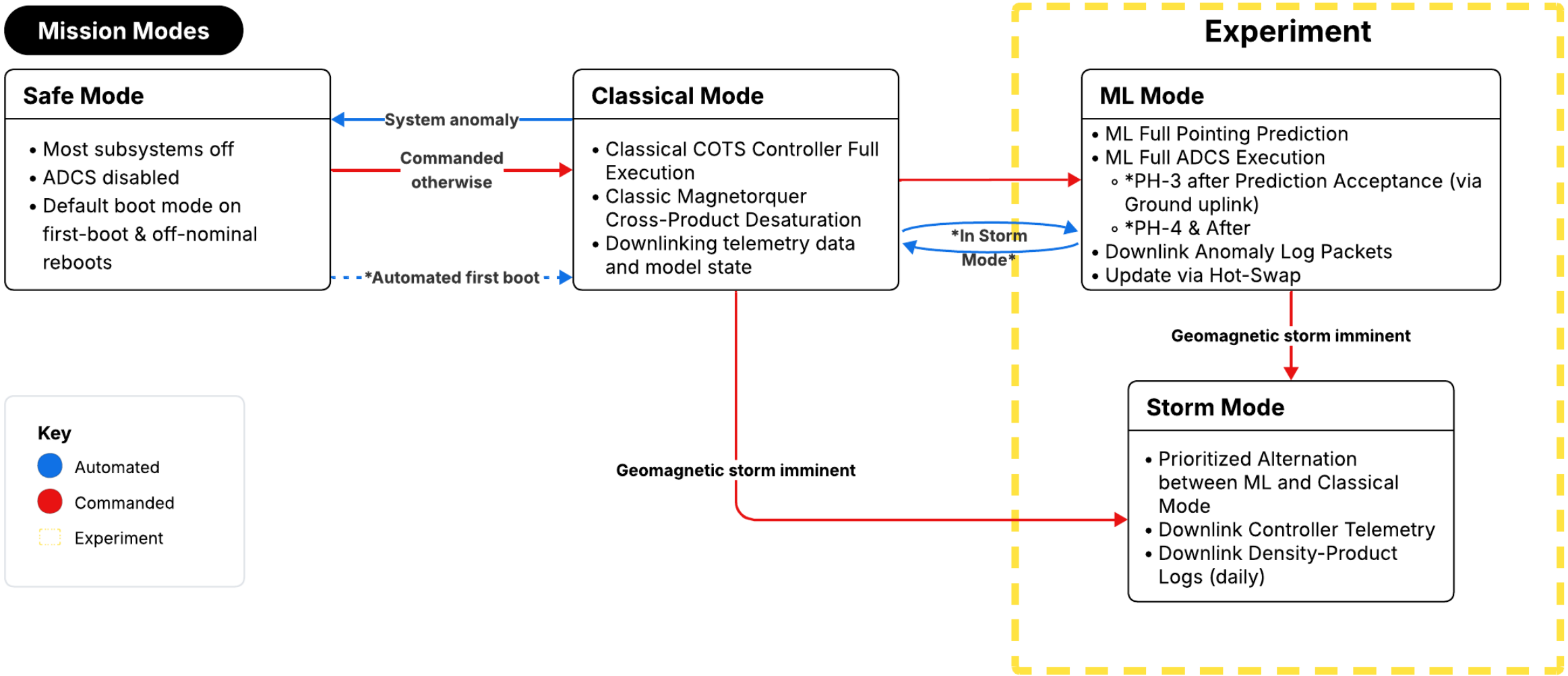
3. Commanded slew to next target

4. Command slew to downlink/uplink
(If uplinking this orbit)

5. Downlink telemetry & any log packets
Uplink model updates/commands

6. Update Model
(If ML Mode & updating this orbit)

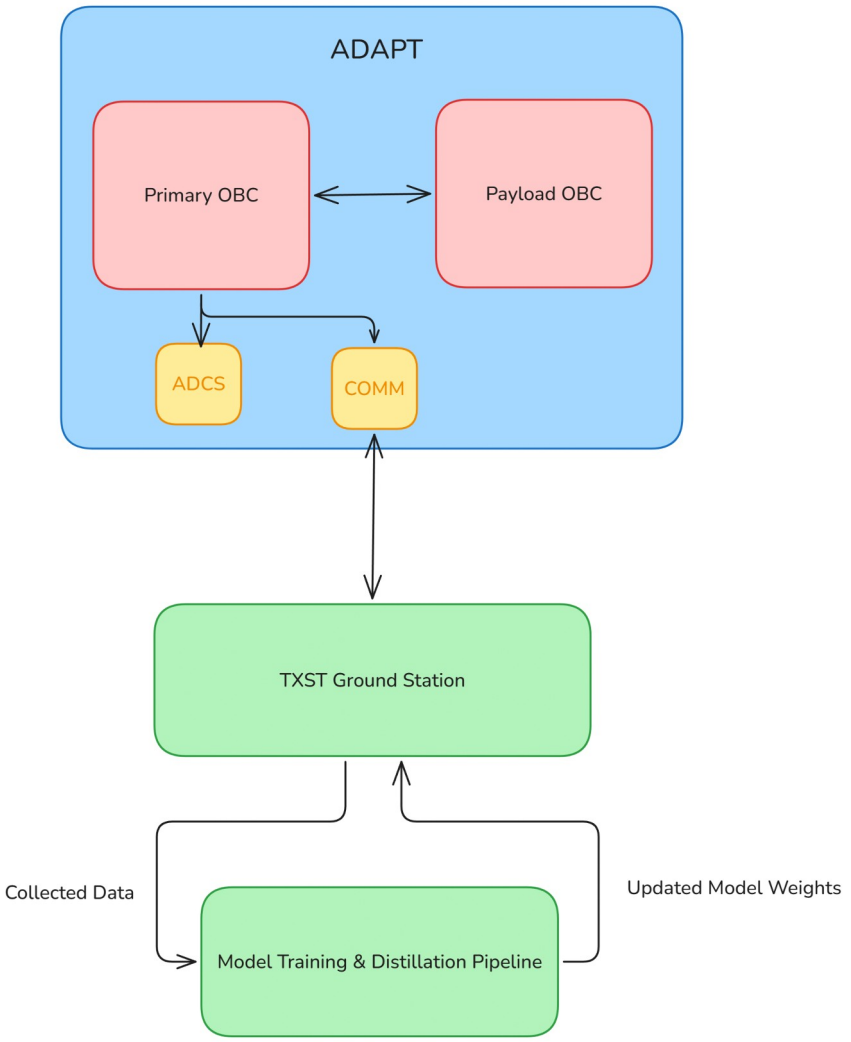
Modes


Mission Modes
Experiment
Safe Mode
• Most subsystems off
• ADCS disabled
• Default boot mode on first-boot & off-nominal reboots
System anomaly
Commanded otherwise
*Automated first boot
Classical Mode
• Classical COTS Controller Full Execution
• Classic Magnetorquer Cross-Product Desaturation
• Downlinking telemetry data and model state
In Storm Mode
ML Mode
• ML Full Pointing Prediction
• ML Full ADCS Execution
◦ *PH-3 after Prediction Acceptance (via Ground uplink)
◦ *PH-4 & After
• Downlink Anomaly Log Packets
• Update via Hot-Swap
Geomagnetic storm imminent
Storm Mode
• Prioritized Alternation between ML and Classical Mode
• Downlink Controller Telemetry
• Downlink Density-Product Logs (daily)
Geomagnetic storm imminent
Key
Automated
Commanded
Experiment

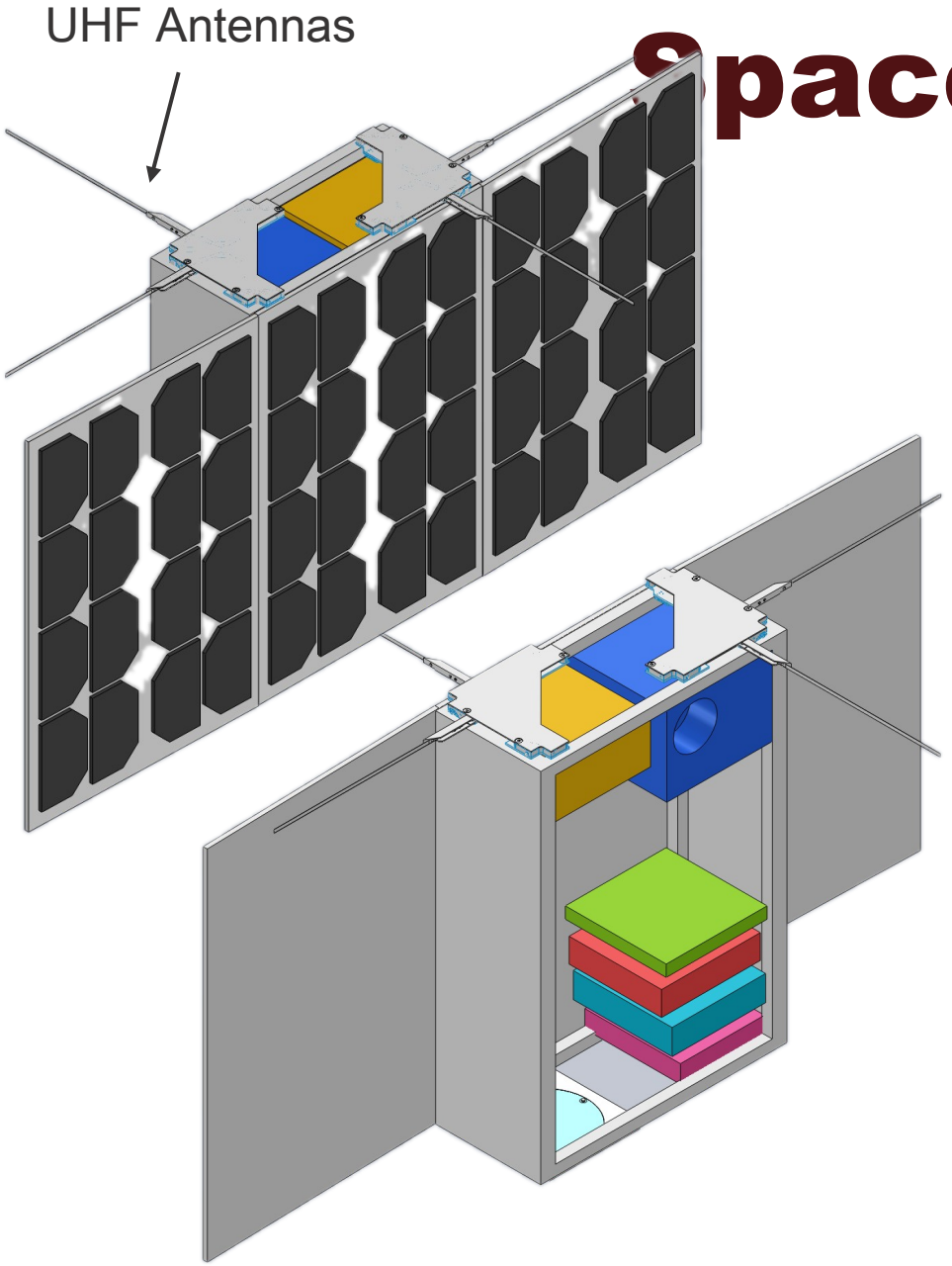
Mission System

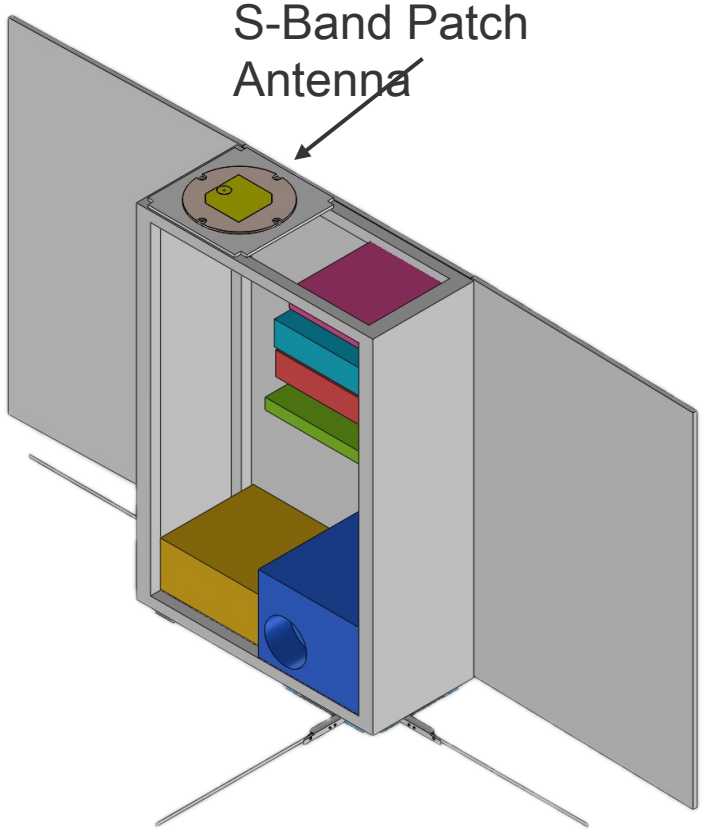
Spacecraft Overview

- 6U CubeSat, deployable solar arrays for power generation
- Dual-OBC structure to eliminate risk of issues with ML model affecting critical flight component functionality.
  - Payload OBC powered off when not in use for power saving.
- Critical subsystem considerations
  - **ADCS** - COTS system, ideally with direct control over actuators for both the classical controller and ML-based controller
  - **FSW** - Handles communication between primary OBC and payload OBC, what ADCS controller is active, and safety checks before executing ML model ADCS commands.
  - **Payload** - Generates ADCS commands, model must be updatable
  - **COMM** - Dual UHF & S-Band radios for increased data throughput
    - S-Band included for now based on expected data throughput requirements but may be removed if unnecessary.



Spacecraft Layout





- XACT 50 ADCS
- SOLO-EPS 4
- EnduroSat OBC
- GPU002SF
- S-Band Transceiver
- UHF Transceiver

Link Budgets

UHF Downlink

UHF Uplink

S-Band Downlink

S-Band Uplink

Power Draw

CDH	EnduroSat OBC w/ builtin GNSS receiver	3.00	100.00%	3.00	100.00%	3.00	100.00%	3.00	100.00%	3.00
			0.00%	0.00	0.00%	0.00	0.00%	0.00	0.00%	0.00
COMM	EnduroSat S-Band Transceiver [Tx]	12.00	0.00%	0.00	25.00%	3.00	100.00%	12.00	0.00%	0.00
	EnduroSat S-Band Transceiver [Rx]	1.20	0.00%	0.00	0.00%	0.00	100.00%	1.20	0.00%	0.00
	EnduroSat RF-Band Transceiver [Tx]	8.00	20.00%	1.60	20.00%	1.60	20.00%	1.60	20.00%	1.60
	EnduroSat RF-Band Transceiver [Rx]	1.00	100.00%	1.00	100.00%	1.00	100.00%	1.00	100.00%	1.00
	VHF/UHF Antenna	0.08	100.00%	0.08	100.00%	0.08	100.00%	0.08	100.00%	0.08
GPS	N/A	0.00	0.00%	0.00	0.00%	0.00	0.00%	0.00	0.00%	0.00
ADCS	ADCS - XACT 50	2.83	0.00%	0.00	100.00%	2.83	100.00%	2.83	100.00%	2.83
			0.00%	0.00	0.00%	0.00	0.00%	0.00	0.00%	0.00
			0.00%	0.00	0.00%	0.00	0.00%	0.00	0.00%	0.00
Power	Power System - SOLO-EPS-4	0.50	100.00%	0.50	100.00%	0.50	100.00%	0.50	100.00%	0.50
	Heaters	2.00	50.00%	1.00	50.00%	1.00	50.00%	1.00	50.00%	1.00
			0.00%	0.00	0.00%	0.00	0.00%	0.00	0.00%	0.00
			0.00%	0.00	0.00%	0.00	0.00%	0.00	0.00%	0.00
Payloads										
Payload	Novo Space GPU002SF/ML Computer	15.00	0.00%	0.00	0.00%	0.00	50.00%	7.50	50.00%	7.50
Total Mode Power (W)			**7.18**		**13.01**		**30.71**		**17.51**	

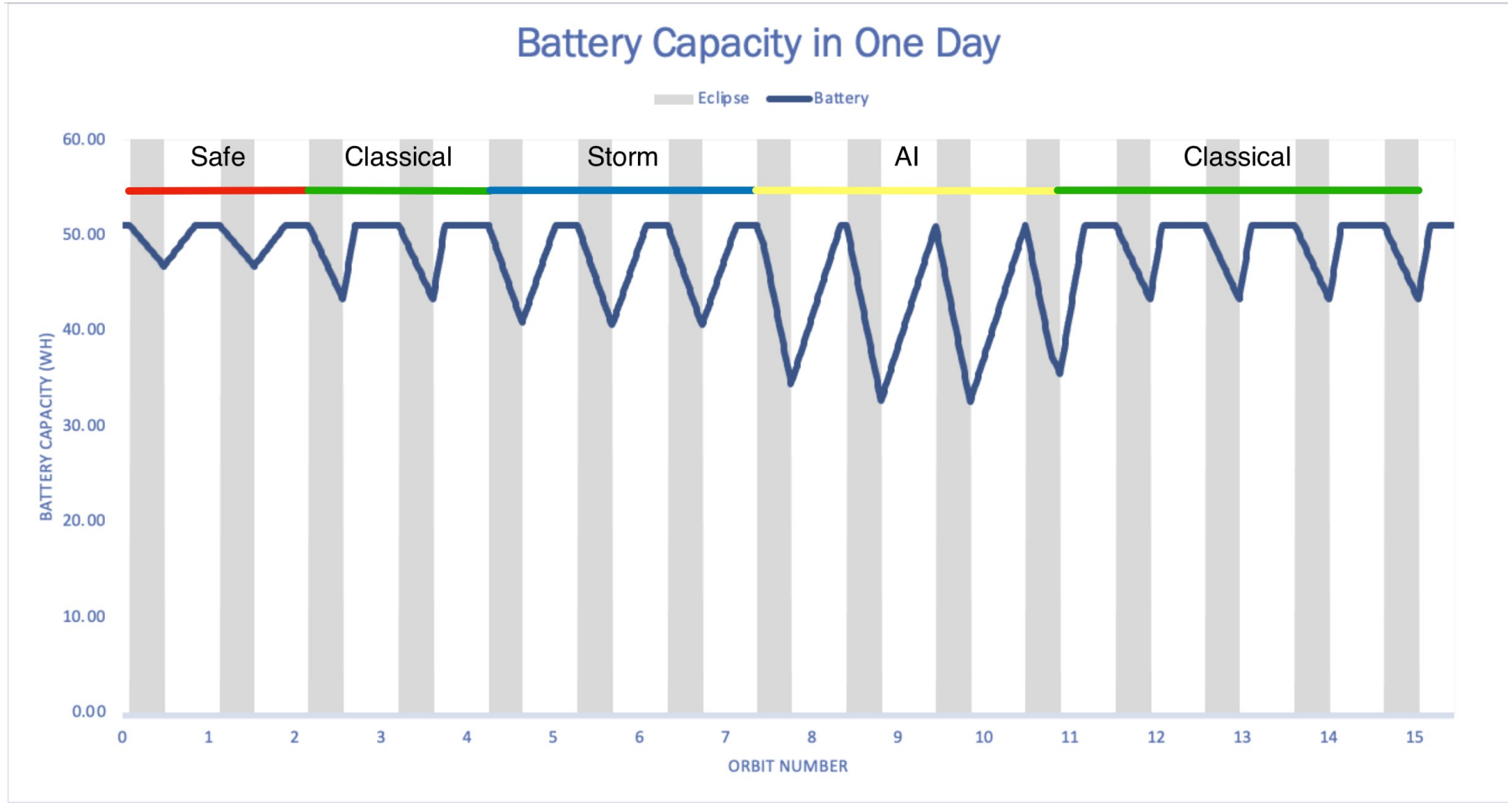
Power Generation

Array	Array Power (W) P_{MP}
Deployable	19.50
Deployable	19.50
Face	19.50
	0.00
	0.00
	0.00
Totals	**N/A**
Total Peak Power (W)	**58.51**

Mode	Power Generated
Safe	15.00
Classical	50.00
AI	50.00
Storm	35.00

Storage (Battery)						
Cell Voltage (V)	Cell Capacity (Ah)	Series Cells/String	# of Strings	Pack Voltage (V)	Pack Capacity (Ah)	Pack Power (Wh)
3.7	3.45	4	1	14.8	3.45	51.06

Power Balance



Data Budget

INPUTS		
Pass Time	10.00	minutes
Number of passes in a day	8	
Radio Data Rate	19,600	bps
Margin	50	%
RESULTS		
Number of passes to downlink all data	3	
Number of days to downlink all data	0	
Total Bytes Generated (1 Day)	**2,263,632**	Bytes
Total Bytes with Margin	**3,395,447**	Bytes
Total (kB)	3,316	kB
Total (MB)	3	MB
Total (GB)	0	GB

UHF Radio

RADIO 2 INPUTS		
Pass Time	10.00	minutes
Number of passes in a day	8	
Radio Data Rate	125,000	bps
Margin	50	%
RESULTS		
Number of passes to downlink all data	1	
Number of days to downlink all data	0	
Total Bytes Generated (1 Day)	**5,704,285**	Bytes
Total Bytes with Margin	**8,556,428**	Bytes
Total (kB)	8,356	kB
Total (MB)	8	MB
Total (GB)	0	GB

S-Band Radio

Mass Budget

MASS BUDGET				
Component	**Material**	**Quantity**	**Mass (kg)**	**Total Mass (kg)**
Structure				**3.08**
2NDSpace CORE-12		3	0.289	0.867
Structure		1	1.1	1.1
Fastners		1	0.189	0.189
Harness		1	0.924	0.924
EPS				**0.48**
2NDSpace SOLO-EPS-4		1	0.48	0.48
ADCS				**1.23**
BCT XACT-50		1	1.23	1.23
COM				**0.57**
EnduroSat S-Band Transceiver		1	0.246	0.246
EnduroSat S-Band Antenna Wideband		1	0.115	0.115
EnduroSat UHF Transceiver		1	0.094	0.094
ISISPACE UHF Antenna		1	0.115	0.115
Payload				**0.14**
NOVO SPACE GPU002SF		1	0.14	0.14
Thermal				**0.05**
Heaters		1	0.05	0.05
CDH				**0.173**
EnduroSat OBC w/ GNSS		1	0.173	0.173
				0
			Total	**5.723**
			Total + 25%	**7.15375**
			Max Allowable	**12**

Subsystem Contribution

Attitude Determination and Control System (ADCS)

- Develop spacecraft attitude determination, pointing, and control capabilities to meet mission performance requirements.
- Close Relation to:
  - Payload

Key Development Areas

- Attitude determination and state estimation
- Controller development and tuning
- Pointing performance characterization

Key Deliverables

- Validated attitude control architecture
- Pointing performance verification
- Baseline controller performance dataset
- ADCS simulation environment

Communications System (COMMS)

- Develop reliable command, telemetry, and data-transfer capabilities between the spacecraft and ground segment.
- Close Relation to:
  - Payload
  - CDH

Key Development Areas

- Communications architecture and link analysis
- Ground segment integration
- Mission communications operations

Key Deliverables

- Communications budget
- Telemetry and command definitions
- Downlink strategy
- Ground segment operations concept

Command and Data Handling (CDH)

- Develop the flight software and onboard computing infrastructure that manages spacecraft operations and mission data.
- Close Relation to:
  - Payload
  - ADCS

Key Development Areas

- Flight software architecture
- Data handling and telemetry management
- Fault protection and mission operations logic

Key Deliverables

- Flight software architecture document
- Mode transition diagram
- Telemetry database
- Fault management framework

Electrical Power System (EPS)

- Develop the spacecraft power generation, storage, and distribution system to support all mission operations.
- Close Relation to:
  - Payload
  - ADCS

Key Development Areas

- Power budgeting and energy management
- Solar array and battery sizing
- Operational power planning

Key Deliverables

- End-to-end power budget
- Energy balance analysis
- Operational power modes
- Battery management strategy

Payload System

Develop and validate the onboard machine learning experiment, including algorithm execution, data collection, and performance assessment.

- Close Relation to:
  - ADCS

Key Development Areas

- ML controller development and validation
- Onboard model execution and experiment operations
- Performance assessment and data analysis

Key Deliverables

- Flight-ready ML controller
- Experiment operations concept
- Performance evaluation framework
- On-orbit experiment dataset

Structures/Thermal System (STR/THERM)

- Develop the mechanical and thermal systems that ensure spacecraft survivability and functionality in launch and space environments.
- Close Relation to:
  - Payload
  - *All systems

Key Development Areas

- Structural design and spacecraft packaging
- Mass and mechanical analysis
- Thermal control and environmental qualification

Key Deliverables

- Mechanical layout and CAD model
- Mass budget
- Structural analysis report
- Thermal control plan

Summary

- Payload: Develop and validate the ML controller.
- ADCS: Measure and optimize control performance.
- CDH: Manage software integration and mission operations.
- Comms: Return experiment data to the ground. EPS and Structures/Thermal: Keep the spacecraft powered, stable, and operational.

Conclusions

Key Takeaways

- The mission's primary goal is to demonstrate an AI-based attitude controller in low earth orbit.
- Success depends on collecting enough data to compare AI control against traditional control methods.
- Software integration and testing will be major development challenges.
- Safety and verification are essential before allowing AI-generated control actions.
- Most engineering effort will focus on integration, validation, and performance evaluation rather than new hardware development.

Thank you!